AL2002
Artificial Intelligence - Lab
Week-1

Objectives:

- To demonstrate the basics of Python Programming Environments ***(Python Interpreter, VS Code, PyCharm, Jupyter Notebook, Google Colab)***, Data Types, Flow Controls, Range Functions, Slice Functions.

Task 1 - Variables & Memory Addresses

1.1: Variable Assignment and Memory Concatenation

Components:

- Variables
- id() function (Memory Address)
- Type Casting (str())
- String Concatenation (+)

Instructions:

- Create a variable named my_var and assign it a value (integer or string).
- Use the id() function to get the memory address of that variable.
- **Critical Step:** Since the address is an integer and the value might be a string (or int), you must convert the address to a string using str() before concatenating.
- Concatenate the value and the address into a single string.
- Print the final result.

Task 2 - Flow Control (Decision Making)

1.2: Simple Grading System using If-Elif-Else

Components:

- input() function
- int() conversion
- if, elif, else statements

Instructions:

- Ask the user to input their marks (0-100).

- Convert the input to an integer.
- Use flow control to determine the grade:
    - 90+: Grade A
    - 80-89: Grade B
    - 70-79: Grade C
    - Below 70: Fail
- Print the assigned grade.

Task 3 - Loops and Range

1.3: Generating Number Sequences

Components:

- for loop
- range(start, stop, step) function

Instructions:

- Write a loop that prints the first 10 numbers (0 to 9).
- Write a second loop using `range()` with a "step" to print only **Even** numbers between 10 and 30.
- Print "Loop Finished" at the end.

Task 4 - Slicing Lists and Strings

1.4: extracting sub-parts of data

Components:

- Lists / Strings
- Slicing Syntax [start:stop]
- Negative Indexing

Instructions:

- Create a string variable: text = "Artificial Intelligence Lab".
- Use slicing to extract just the word "Intelligence".
- Create a list of numbers: data = [10, 20, 30, 40, 50, 60].
- Use slicing to print the last three items using negative indexing.
- Use slicing to reverse the list.